

# New Autism Diagnostic Interview-Revised Algorithms for Toddlers and Young Preschoolers from 12 to 47 Months of Age

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**Abstract** Autism Diagnostic Interview-Revised (Rutter et al. in Autism diagnostic interview-revised. Western Psychological Services, Los Angeles, 2003) diagnostic algorithms specific to toddlers and young preschoolers were created using 829 assessments of children aged from 12 to 47 months with ASD, nonspectrum disorders, and typical development. The participants were divided into three more homogeneous groups by language level and age. Items that best differentiated the diagnostic groups were selected and arranged into domains based on multi-factor item-response analyses. Using the new algorithms for toddlers and preschool children, we were able to improve sensitivity and specificity compared to the previously developed algorithm.

**Keywords** Autism · Autism spectrum disorders · ADI-R · Toddler · Preschool children · Diagnosis

## Introduction

The Autism Diagnostic Interview-Revised (ADI-R, Lord et al. 1994) is a standardized, semistructured, investigator-based interview for parents or caregivers of individuals referred for a possible Autism Spectrum Disorder (ASD). The ADI-R includes 93 items in three domains of functioning—language/communication; reciprocal social interactions; and restricted, repetitive, and stereotyped behaviors and interests, as well as other aspects of

behaviors. Up to 42 of the interview items are systematically combined to produce a formal, diagnostic algorithm for autism (Rutter et al. 2003) based on the ICD-10 (World Health Organization 1990) and DSM-IV (American Psychiatric Association; APA 1994) definitions of autism as specified by the authors. Other criteria such as using lower cutoffs with the same set of items have been used to create an algorithm for broader classification of autism spectrum disorders (ASD) as used in several collaborative studies (Dawson et al. 2004; Risi et al. 2006). Previous analyses suggested that the diagnostic algorithm was useful for children with a non-verbal mental age above 2 years (Lord et al. 1994). Because most toddlers and preschool children with ASD are not yet at this level of skill, the ADI-R algorithm has not been appropriate to characterize very young children with severe delays (Ventola et al. 2006).

A ‘Toddler’ version of the ADI-R was developed several years ago to provide descriptive data to be used for research purposes with children under 4 years of age. It includes 32 new questions and codings about the onset of autism symptoms and general development with a total 125 items. Other items in both versions of the ADI-R are identical except that the Toddler ADI-R does not have codes for behaviors between 4 and 5 years of age (referred to as *most abnormal 4–5*). No diagnostic algorithm was generated for the toddler version of the ADI-R.

Because of the belief that earlier provision of services and treatments is associated with better outcomes, in the past few years, research has flourished concerning detection of ASD symptoms in the first 2 years of life (National Research Council 2001). In recent studies, the average age of first parental concern was between 15 and 18 months (Chawarska et al. 2007; DeGiacomo and Fombonne 1998). Advocacy and funding agencies have also joined together to promote the study of infant siblings of children with

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autism and other very young children at risk for ASD as seen in the establishment of the Baby Siblings Research Consortium (Yirmiya and Ozonoff 2007). Thus, researchers and clinicians recognize the increasing need for diagnostic measures that are appropriate for toddlers to capture the early signs of autism at such young ages. For example, the Toddler ADI-R was used to study the parental recognition of developmental problems in toddlers with ASD (Chawarska et al. 2007). Lord et al. (2004) examined regression and word loss in toddlers with ASD using the Toddler ADI. Another study focused on restricted and repetitive behaviors in young children with ASD based on the Toddler ADI-R in addition to other measures (Richler et al. 2007). The purpose of the present study is to propose a first set of diagnostic algorithms for toddlers and young preschool children developed on a sample of children whose ages ranged from 12 to 47 months with nonverbal mental ages down to 10 months. Although the initial intent was to refine the existing toddler ADI-R into a specific instrument for toddlers, in creating algorithms specifically for young children, priority was given to items that overlapped between the toddler and standard versions of the ADI-R because of the wider availability of the standard ADI-R.

The published algorithms for the ADI-R include a *current behavior algorithm* form (as distinguished from an empirically supported diagnostic algorithm) for children whose ages range from 2 years, 0 months to 3 years, 11 months. Age 4 is a natural dividing point because the standard ADI-R contains questions about children's behavior between age of 4 and 5 (48–59 months) that are not applicable to younger children. The list of items on this form has been used to describe toddlers whose caregivers were administered either the Toddler or standard version of the ADI-R (Wiggins and Robins 2008). However, sensitivity and specificity of this list of items with very young children have not yet been carefully examined. In fact, the study that provided the psychometric properties of the existing ADI-R algorithms was based on a sample of children from 36 to 59 months of age, with mental ages ranging from 21 to 74 months (Lord et al. 1994). Using the existing algorithms, the group of children with autism over 36 months of chronological age was well differentiated from children with nonspectrum disorders showing high sensitivity and specificity (both over 0.90). Further analyses of data from preschoolers revealed that the ADI-R algorithms significantly differentiated between children over 2 years with ASD from other developmental disorders. However, discrimination between nonverbal children with ASD and nonverbal children without ASD under 2 years of age, especially for those with mental ages under 18 months was poor, resulting in low specificity (Lord et al. 1993). Analyzing a larger sample, Risi et al. (2006)

also showed high sensitivity (above 80%) of the ADI-R for the classification of children with ASD under 3 years of age, but lower specificity for these children in the comparison of ASD versus nonspectrum disorders (around 70%).

In contrast, Ventola et al. (2006) reported that the algorithm for the ADI-R resulted in lower sensitivity when compared with Autism Diagnostic Observation Schedule (ADOS; Lord et al. 2000), Childhood Autism Rating Scale (CARS; Schopler et al. 1980), and clinical judgment using DSM-IV criteria for children whose chronological age ranged from 16 to 30 months. In the study by Ventola et al. (2006), parents did not report sufficient Repetitive and Restrictive Behaviors (RRBs) to meet the criteria for the RRB domain on the ADI-R. Wiggins and Robins (2008) also found that ADI-R algorithms resulted in poor sensitivity for children between 16 and 37 months of age when the standard cutoff for the RRB domain was included in the diagnostic criteria. The authors suggested that, although RRBs are common in children with ASD, the RRB domain is not as relevant to the ADI-R classification as other diagnostic criteria when evaluating very young children since these behaviors are less often reported by their parents. Thus, despite the relatively good sensitivity and specificity of the ADI-R diagnostic algorithms for older preschool and school age children, there is a need to develop diagnostic algorithms with improved sensitivity and specificity that can differentiate toddlers and young preschoolers with ASD from children with nonspectrum disorders.

Recently, large datasets from several collaborative efforts were used to generate improved algorithms for another autism diagnostic instrument, the ADOS (Lord et al. 2000). In the course of these analyses, it was found that the use of different algorithms for subsets of children defined by age and language level reduced the effects of age, language level, and IQ and increased the sensitivity and specificity of the measure (Gotham et al. 2007). Consequently, we used a similar strategy in developing the new ADI-R algorithms for toddlers and young preschoolers. In addition, because recent studies have shown that classifications within ASD, that is, diagnoses of autism versus Pervasive Developmental Disorder-Not Otherwise Specified (PDD-NOS) are relatively unstable in contrast to overall diagnoses of ASD versus other nonspectrum disorders that are consistent over time (Lord et al. 2006; Turner and Stone 2007; Kleinman et al. 2008) and in keeping with the proposed DSM-V ASD criteria ([www.dsm5.org](http://www.dsm5.org)), the new algorithms include only two classifications: ASD or nonspectrum.

In order to formally acknowledge the less clear stability of diagnoses in younger children, as we had for the ADOS-Toddler Module (ADOS-T; Lord et al., in press; Luyster

et al. 2009), in the present paper, we propose *ranges of concerns* to be used as the primary outcome of the algorithms following the structure of the ADOS-T algorithms. The ranges can be used by clinicians for clinical monitoring and follow-up (rather than yes/no cutoffs for ASD), reflecting little-to-no, mild-to-moderate, or moderate-to-severe concern. However, reflecting the necessity in some research and clinical settings for more strictly stratified groupings, two cutoff scores, one for research (more restrictive; higher specificity with lower sensitivity) and one for clinical purposes (more inclusive; higher sensitivity with lower specificity), were selected for each algorithm. These alternatives allow researchers and clinicians to be transparent about the choices they make, recognizing that diagnostic decisions about ASD in very young children are less stable and precise than for older children and adolescents.

As with other diagnostic measures, the algorithm scores are not intended to be used alone in generating a diagnosis. The assumption is that both clinicians and researchers will make diagnostic decisions taking into account children's developmental history, clinical observations, cognitive skills, and social, language and adaptive functioning in various contexts (Lord and Corsello 2005). The ADI-R can be also used to describe characteristics of ASD in individual children. Theoretically generated and empirically derived factors are used to organize autism symptoms into different domains. Reliability at the item level allows researchers the opportunity to develop different constructs to study behavioral differences, as well as clinicians the opportunity to describe a child's strengths and weaknesses.

In order to examine the domain structure of autism symptoms included in the ADI-R for children under 4 years of age, we performed factor analyses using data obtained from toddlers to preschoolers. The existing diagnostic algorithms for the ADI-R capture abnormalities in 3 symptom domains, *Reciprocal Social Interaction, Communication, and Restricted and Repetitive Behaviors*, which is consistent with the current framework of the DSM-IV. Several recent studies have found other organizations of autism symptom domains in the ADI-R. The content areas that comprise these domains have varied across different samples and analytic techniques. One study found two factors, *social and communication* and *restricted and repetitive behaviors* using confirmatory and exploratory factor analyses based on the data from the ADI-R for children from 4 to 18 years of age (Snow et al. 2008). Another study found that a three-factor solution (*peer interaction and imaginative play, social and communication, and stereotyped behavior/restricted interests*) fit better than one- or two-factor solutions based on the data from the ADI-R for children and adolescents from 4 to 20 years of age (Van Lang et al. 2006). The present paper uses data

from toddlers to young preschoolers to replicate the results of the past studies. Factor analyses are used to identify new algorithm domain scores that can represent patterns of autism symptoms as well as to generate guidelines for combining domain scores in ways that maximize sensitivity and specificity.

## Methods

### Participants

Analyses were conducted on 695 different *participants*. Some participants had repeated assessments yielding a total of 829 *cases* (each case was defined by complete data of an ADI-R, contemporaneous ADOS, nonverbal IQ, and best estimate clinical diagnosis). The majority of participants were recruited from two projects, Early Diagnosis of Autism (EDX) and First Words and Toddlers (FW/T) at the University of Michigan Autism and Communication Disorders Center (UMACC) and University of Chicago Developmental Disorders Clinic. The rest of participants were school- or physician-referred clinic patients at UMACC. Children in the EDX project were assessed at ages 2 and 3. Children in the FW/T were assessed at entry to the study (approximately at 2 years of age) and at 3.

The sample was limited to participants aged 12–47 months with a nonverbal mental age of at least 10 months. Out of 695 participants, 535 were males. The mean age of the participants was around 33 months (Mean ( $M$ ) = 33.3, Standard Deviation (SD) = 9.4). The dataset included 491 participants with ASD, 136 participants with non-spectrum disorders (NS), and 67 typically developing participants. Non-spectrum participants had a range of diagnoses, including language disorders (37%), nonspecific developmental delays (cases with either the nonverbal or verbal IQ scores more than one standard deviation below average; 21%), non-specific intellectual disability (cases with the full scale IQ score less than 70 with significant delays in adaptive skills; 16%), and behavioral disorders (8%). Ethnicity was not associated with diagnosis; 78% of participants were Caucasian, 16% African American, 2.5% Asian American, 2.5% biracial, and about 1% of participants were Native Americans or others.

We created three developmental cells to obtain more homogeneous groups to reduce the effect of language level and age. These groups were defined by age and verbal status during the assessment. The “phrase speech (PH)” group included children whose scores on the item in the ADI-R, “Overall Level of Language” were ‘0’ = spontaneous language that involves phrases of three words or more including verbs. The “single words (SW)” group included children who scored ‘1’ = use of speech that

involves at least five different spontaneous, meaningful words used daily. The “nonverbal (NV)” group included children whose scores on this item were either ‘2’ = fewer than five words total used on a daily basis or ‘3’ = no words used in a daily basis. Consistent with the past finding by Luyster et al. (2009) based on the ADOS-T scores, we found that the ADI-R score distributions of younger children (from 12 to 20 months, 30 days) and older nonverbal children (from 21 to 47 months, 30 days) were similar to each other. In contrast, children 21 months or older who had single words or phrase speech differed in the score distributions. Thus, the developmental groups were assigned as follows: (1) all children 12–20 months, 30 days of age as well as nonverbal children 21–47 months, 30 days of age (hereafter referred to as “12–20/NV21–47”); (2) children 21–47 months, 30 days of age with single words (“SW21–47”); (3) all children 21–47 months, 30 days of age with phrase speech (“PH21–47”).

Some analyses (factor analyses, logistic regression analyses) were run for data from all 829 assessments (referred to as “*all cases*”) in order to take advantage of the larger sample size afforded by including repeated measurements. However, in order to eliminate effects of repeated participation on domain totals and algorithm totals, the majority of analyses (ROC curves, correlations, and internal consistency) were run with “*unique cases*,” a dataset in which each participant was represented only once per group. Mean ages, NVIQ scores, and gender distributions varied by diagnosis and developmental cells for both *all cases* and *unique cases*. In the *unique cases* groups, the results from the one-way ANOVA indicated that nonverbal IQ differed significantly across three diagnostic groups for “12–20/NV21–47” and “PH21–47” groups with typically developing children showing the highest mean NVIQ scores ( $p < 0.001$ ). Typically developing children were younger than children with ASD and NS ( $p < 0.001$ ). Therefore, we controlled for age and IQ scores in all analyses. As expected, the nonverbal group of children between 21 and 47 months of age only included children with ASD and NS because there were no nonverbal typically developing children at this age. Similar patterns were examined for the *all cases* sample. See Table 1 for more details on mean ages, NVIQ scores, and gender distributions by diagnosis and developmental cells for *all cases*.

## Measures

Either the toddler or standard version of the ADI-R was administered by a clinical psychologist or a trainee who had completed research training and met standard requirements for research reliability (See Risi et al. 2006). A developmental hierarchy of psychometric measures, most frequently the Mullen Scales of Early Learning

(MSEL; Mullen 1995; 52% of participants received the MSEL) and the Differential Abilities Scale (DAS; Elliott 1990) were used to determine IQ scores. For the MSEL, ratio IQ scores were used as estimates of ability in the present study as in many past studies (e.g. Richler et al. 2007) based on the finding that the ratio IQ scores derived from the MSEL using the age equivalents were found to have a good convergent validity with the DAS (Bishop et al., in press). The MSEL ratio IQs were calculated by averaging the age equivalents of all subtests to obtain mental age, and then dividing mental age by chronological age and multiplying by 100. Cognitive testing generally took place immediately before the ADOS administration within the same week as the ADI-R. The ADOS-T (Lord et al., in press), Module 1 or 2 of the ADOS (Lord et al. 1999), or Pre-Linguistic ADOS (PL-ADOS; DiLavore et al. 1995) was administered depending on the age and developmental level of the child. Out of 829 cases, 294 cases were administered the PL-ADOS because they were assessed before the ADOS modules 1 and 2 were developed.

## A Consensus Best Estimate Diagnosis

For children in the EDX study, an experienced clinical researcher used the ADOS and ADI-R scores and observations made during the testing to generate independent best estimate diagnoses of autism, PDD-NOS, and non-spectrum disorders (APA 1994; See Lord et al. 2006). For children in the FW/T project, scores on the ADI-R, ADOS, and clinical observations were used by two clinicians to make a best estimate diagnosis operationalizing DSM-IV criteria for autism and PDD-NOS (APA 1994; See Luyster et al. 2009). For clinic cases, a diagnosis was made by a psychologist and/or psychiatrist after review of all data mentioned above.

## Design and Analyses

First, we replicated the previously developed algorithm in order to be able to compare the predictive validity obtained from the previous algorithm to the new algorithms. For the development of the new algorithms, the first step was to examine distributions for all of the items in toddler and standard version of ADI-R and identify those that best differentiated ASD and NS diagnoses (referred to as “best items”). Most items in the ADI-R are coded with *no definite behavior of the type specified* (score of 0), *behavior of the type specified probably present but defining criteria not fully met* (score of 1), and *definite abnormal behavior of the type described in the definition and coding* (score of 2), with a score of 3 used occasionally to indicate *extreme severity* (Rutter et al. 2003). Scores of 2 and 3 are collapsed

**Table 1** Description of sample of *all cases*

|                                 | 12-20         |               |                | NV21-47          |                | SW21-47          |                | PH21-47         |                |               |
|---------------------------------|---------------|---------------|----------------|------------------|----------------|------------------|----------------|-----------------|----------------|---------------|
|                                 | ASD           | NS            | TD             | ASD              | NS             | ASD              | NS             | ASD             | NS             | TD            |
| <i>N (male, female)</i>         | 43<br>(37, 6) | 24<br>(16, 8) | 47<br>(30, 17) | 318<br>(264, 54) | 36<br>(20, 16) | 140<br>(103, 37) | 32<br>(19, 13) | 107<br>(91, 16) | 62<br>(49, 13) | 20<br>(13, 7) |
| <i>Chronological age</i>        |               |               |                |                  |                |                  |                |                 |                |               |
| Mean                            | 17.5          | 16.7          | 15.6           | 34.2             | 29.8           | 36               | 33.6           | 40.9            | 38.7           | 28.4          |
| Standard deviation              | 2.2           | 2.4           | 2.6            | 7.2              | 6.4            | 6.4              | 5.9            | 5.4             | 5.7            | 5.2           |
| Minimum–maximum                 | 13–20         | 12–20         | 12–20          | 21–47            | 22–43          | 21–47            | 22–46          | 24–47           | 22–47          | 21–37         |
| <i>Nonverbal IQ<sup>a</sup></i> |               |               |                |                  |                |                  |                |                 |                |               |
| Mean                            | 85.3          | 86.4          | 111            | 60.7             | 64.9           | 71.4             | 69.1           | 87.2            | 92.1           | 121           |
| Standard deviation              | 20.2          | 23.9          | 16.6           | 20.7             | 30.9           | 18.8             | 18.1           | 22.2            | 17.9           | 17.8          |
| Minimum–maximum                 | 41–119        | 38–123        | 68–150         | 15–144           | 17–132         | 32–122           | 47–114         | 31–156          | 50–129         | 83–155        |

<sup>a</sup> Full scale scores were used for 39 children with TD in the 12-20 group, 1 child with NS in the SW21-47 group, 1 child with ASD, 2 children with NS, and 15 children with TD in the PH21-47 group because no nonverbal scale was available

*12-20* Children from 12 to 20 months of age, *NV21-47* Nonverbal children from 21 to 47 months of age, *SW21-47* Children with single words from 21 to 47 months of age, *PH21-47* Children with phrase speech from 21 to 47 months of age, *ASD* autism spectrum disorder, *NS* nonspectrum disorder, *TD* typical development

to “2” in the algorithm. The intended selection criteria for the best items were that no more than 20% of ASD cases scored a zero on an item, and no more than 20% of non-spectrum cases scored a 2 or 3. For some items that were theoretically important, these criteria were modified to include items with slightly less clear distributions for certain developmental cells (i.e. *hand and finger mannerisms, compulsions and rituals* for “PH21-47” group). ANOVAs were also used to confirm the results of the diagnostic differences. Items that did not differentiate between diagnoses according to the ANOVAs were excluded from the final sets. Correlations were generated between items and participant characteristics such as age and IQ separately for each developmental cell, and items that were highly correlated with participant characteristics were not included in the final set of items. In addition, when items were highly correlated with each other (above an  $r$  of 0.5), the item that better differentiated ASD versus NS was selected.

After we selected the items that best differentiated between diagnoses, we performed exploratory factor analyses separately for each of the three developmental cells. Scores of 2 and 3 were collapsed and scores of 7, 8 and 9 were treated as missing data and excluded. Because the ADI-R data are ordinal and scores do not represent equal intervals, analyses were run as ordinal probit item response models using Mplus Version 5.21 software. Geomin oblique rotations were used to examine the pattern of factor loadings, and some items were added and deleted based on the loadings to maximize the model fit. Confirmatory factor analyses were next used to verify goodness-of-fit for the factor structure. Then, we used logistic regression to examine the weighting of the three domains

created from factor analyses in view of the relative predictive value of scores from the different factors. Item correlations for each individual item with domain-totals-minus-the-item and the correlations between the domains and participant characteristics were examined. ROC curves were calculated to examine the sensitivity and specificity of the selected cutoff scores.

## Results

### Replication of Sensitivity and Specificity of the Previous Algorithm

We first used ROC curves (Siegel et al. 1989) to examine sensitivity and specificity of the previous *current behavior algorithm* in order to compare the predictive validity of the new diagnostic algorithms to application of the previous algorithm (Rutter et al. 2003). For the younger or non-verbal group (“12-20/NV21-47”), the previous algorithm yielded excellent sensitivity (97%) but very poor specificity (43%) for ASD versus NS. When typically developing children were combined with the NS group, specificity was 47%.

For the SW21-47 group, the original algorithm resulted in 91% sensitivity and 82% specificity when children with ASD were compared to NS. For the PH21-47 group, the existing *current behavioral algorithm* showed 70% sensitivity and 68% specificity for the comparison of ASD versus NS. Specificity improved slightly to 70% when the TD group was included. Thus, the biggest challenges in discriminating children with ASD were the youngest

children, nonverbal children with the most significant delays (“12-20/NV21-47”), and preschool children with more advanced verbal abilities (“PH21-47”).

### Development of the New Algorithms

#### Best Items

Items were selected using the criteria mentioned above for each of the three developmental cells defined by age ranges and language level. Items from both the standard and toddler ADI-R were considered. Initial analyses determined that items that appeared in both measures were consistently more informative than items in only one of the measures. Thus, further analyses were restricted to items used in both the standard and toddler versions. All items in the ADI-R are coded in terms of whether the behavior is “currently” occurring (behavior occurred within the past 3 months; referred as the “current” item), and whether it “ever” occurred (referred as the “ever” item, which also includes current behavior). Selected items included a combination of *current* and *ever* items.

#### Factor Analyses

Results from the factor analyses showed that 3-factor solutions fitted well for all three developmental cells (Table 2). For the “12-20/NV21-47” and “SW21-47” groups, items loaded onto three factors, Social Affect (SA), Restricted and Repetitive Behaviors (RRB), and Imitation, Gestures, and Play (IGP). For the “PH21-47” group, items associated with SA and IGP loaded onto a first factor, also referred to as “Social Communication (SC),” and items associated with RRBs comprised a second factor. A third factor emerged for the “PH21-47” group comprised of items related to “Reciprocal and Peer Interaction (RPI).” In most cases, a loading on the exploratory factor analyses (EFA) of 0.4 was set as a threshold for the inclusion criteria to include in the algorithms except for several items that were theoretically important. These items had high specificity though they were not common across all diagnostic groups (e.g. *ever: unusual preoccupation* for the “SW21-47” algorithm, *current: use of other’s body to communicate* for the “PH21-47” algorithm; See Table 2 for more details). Although the loadings for these items were lower than the threshold in the EFA, confirmatory factor analyses showed notably higher loadings. All of the RRB items selected were *ever* items because *ever* RRB items consistently showed higher sensitivity and specificity than *current* RRB items. The one exception was *current: stereotyped language* for children with phrase speech, which differentiated diagnoses better than *ever: stereotyped language*, consistent with the results from the past

study using the original ADI-R (Rutter et al. 2003). All non-RRB items in the proposed algorithms are *current* items.

Confirmatory factor analyses (CFA) were performed to examine the model fit for each group, and the results showed a 3-factor model to fit substantially better than 1-, and 2-factor models for all three groups. In the “12-20/NV21-47” group, the goodness-of-fit rating yielded a Comparative Fit Index (CFI) of 0.952 (CFI between 0.9 and 1 indicating good fit; Skrandal and Rabe-Hesketh 2004) and Root Mean Square Error Approximation (RMSEA) of 0.069 (RMSEA of 0.08 or less is considered a satisfactory fit; Browne and Cudeck 1993).

For the “SW21-47” group, two items, *current: interest in other children* and *current: response to approach of other children*, were not initially included in the EFA. However, because these items emerged in both other algorithms, they were added in the CFA to the first factor, SA, to maximize the number of algorithm items directly comparable across the three age by language groupings. The addition of these items in the first factor also increased the predictive validity of the algorithm resulting in higher sensitivity and specificity. The model fit from the confirmatory factor analyses showed satisfactory fit even after these changes were made (CFI of 0.943 and RMSEA of 0.062).

For the “PH21-47” group, the goodness-of-fit rating was satisfactory with CFI of 0.952 and RMSEA of 0.057. In addition, for all three groups, the item, *current: inappropriate facial expression*, always loaded onto the RRB domain; however, we placed the item into the first domain of the algorithm because it made more sense theoretically.

#### Logistic Regression Check on the Prediction of Diagnosis for Each Domain

In order to describe each child’s individual profile, item scores are totaled for each domain. Item scores that were most predictive of diagnosis based on logistic regressions are then combined to yield a single cutoff for categorical classification of ASD. Because factor scores were not uniformly better at prediction of diagnosis than simple totals for each domain, simple item totals for each factor were used in logistic regression analyses, predicting clinical diagnoses of ASD as the outcome. Logistic regression for ASD versus nonspectrum disorders (NS) indicated that among the domains, the SA domain significantly predicted diagnosis when IQ scores, chronological age, and the two other factors were controlled for the younger and older nonverbal group and the older children with single words ( $\beta = 0.17$ ,  $p < 0.001$  for “12-20/NV21-47”;  $\beta = 0.402$ ,  $p < 0.001$  for “SW21-47”). The RRB domain was also a significant predictor of diagnosis while controlling for the

**Table 2** Algorithm mapping for groups defined by chronological age and expressive language level

| 12-20/NV21-47                          | Factor Loadings |       | SW21-47                                              | Factor loadings |       | PH21-47                                | Factor loadings |       |
|----------------------------------------|-----------------|-------|------------------------------------------------------|-----------------|-------|----------------------------------------|-----------------|-------|
|                                        | EFA             | CFA   |                                                      | EFA             | CFA   |                                        | EFA             | CFA   |
| Social affect                          |                 |       | Social affect                                        |                 |       | Social communication                   |                 |       |
| C. Attention to Voice*                 | 0.69            | 0.79  | C. Attention to Voice*                               | 0.75            | 0.74  | C. Attention to Voice*                 | 0.71            | 0.81  |
| C. Direct Gaze*                        | 0.76            | 0.77  | C. Direct Gaze*                                      | 0.85            | 0.76  | C. Direct Gaze*                        | 0.79            | 0.74  |
| C. Social Smiling†                     | 0.95            | 0.84  | C. Social Smiling†                                   | 0.87            | 0.75  | C. Nodding to mean yes                 | 0.58            | 0.6   |
| C. Seeking to Share Enjoyment*         | 0.47            | 0.85  | C. Seeking to Share Enjoyment*                       | 0.78            | 0.71  | C. Seeking to Share Enjoyment*         | 0.82            | 0.71  |
| C. Range of Facial Expression*         | 0.56            | 0.69  | C. Range of Facial Expression*                       | 0.63            | 0.72  | C. Range of Facial Expression*         | 0.51            | 0.59  |
| C. Inappropriate Facial Expression†    | 0.58            | 0.64  | C. Inappropriate Facial Expression†                  | 0.41            | 0.64  | C. Offers Comfort                      | 0.43            | 0.61  |
| C. Appropriateness of Social Response* | 0.87            | 0.84  | C. Appropriateness of Social Response*               | 0.68            | 0.69  | C. Pointing to Express Interest*       | 0.59            | 0.71  |
| C. Interest in Children*               | 0.91            | 0.81  | C. Interest in Children <sup>a</sup> *               | –               | 0.76  | C. Showing and Directing attention*    | 0.77            | 0.76  |
| C. Response to Approaches of Children* | 0.93            | 0.77  | C. Response to Approaches of Children <sup>a</sup> * | –               | 0.81  | C. Quality of Social Overtures†        | 0.80            | 0.76  |
|                                        |                 |       | C. Quality of Social Overtures†                      | 0.71            | 0.72  | C. Social Chat                         | 0.57            | 0.73  |
|                                        |                 |       |                                                      |                 |       | C. Use of Other's Body to Communicate  | 0.32            | 0.3   |
| Repetitive & Restricted Behaviors      |                 |       | Repetitive & Restricted Behaviors                    |                 |       | Repetitive & Restricted Behaviors      |                 |       |
| E. Repetitive Use of Objects*          | 0.55            | 0.8   | E. Repetitive Use of Objects*                        | 0.53            | 0.76  | C. Stereotyped Language*               | 0.5             | 0.84  |
| E. Hand and Finger Mannerisms *        | 0.46            | 0.67  | E. Hand and Finger Mannerisms *                      | 0.67            | 0.53  | E. Hand and Finger Mannerisms *        | 0.6             | 0.47  |
| E. Other Complex Mannerisms*           | 0.55            | 0.74  | E. Other Complex Mannerisms*                         | 0.64            | 0.63  | E. Other Complex Mannerisms*           | 0.41            | 0.61  |
| E. Unusual Sensory Interests*          | 0.65            | 0.72  | E. Unusual Sensory Interests*                        | 0.34            | 0.65  | E. Unusual Sensory Interests*          | 0.53            | 0.71  |
|                                        |                 |       | E. Unusual Preoccupations†                           | 0.25            | 0.38  | E. Unusual Preoccupations†             | 0.43            | 0.62  |
|                                        |                 |       | E. Compulsions/Rituals†                              | 0.44            | 0.44  | E. Compulsions/Rituals†                | 0.49            | 0.64  |
| Imitation, Gestures & Play             |                 |       | Imitation, Gestures & Play                           |                 |       | Reciprocal and Peer Interaction        |                 |       |
| C. Pointing to Express Interest*       | 0.72            | 0.86  | C. Pointing to Express Interest*                     | 0.59            | 0.71  | C. Appropriateness of Social Response* | 0.5             | 0.89  |
| C. Conventional/Instrumental Gestures† | 0.68            | 0.88  | C. Conventional/Instrumental Gestures†               | 0.69            | 0.79  | C. Interest in Children*               | 0.86            | 0.84  |
| C. Spontaneous Imitation of Actions†   | 0.77            | 0.83  | C. Spontaneous Imitation of Actions†                 | 0.84            | 0.78  | C. Response to Approaches of Children* | 0.73            | 0.8   |
| C. Offering to Share†                  | 0.71            | 0.83  | C. Offering to Share†                                | 0.5             | 0.62  |                                        |                 |       |
| C. Imaginative Play†                   | 0.82            | 0.69  | C. Imaginative Play†                                 | 0.58            | 0.58  |                                        |                 |       |
| C. Showing and Directing Attention*    | 0.63            | 0.89  |                                                      |                 |       |                                        |                 |       |
|                                        | EFA             | CFA   |                                                      | EFA             | CFA   |                                        | EFA             | CFA   |
|                                        | CFI             |       |                                                      | CFI             |       |                                        | CFI             |       |
|                                        | 0.991           | 0.952 |                                                      | 0.988           | 0.943 |                                        | 0.988           | 0.96  |
|                                        | RMSEA           |       |                                                      | RMSEA           |       |                                        | RMSEA           |       |
|                                        | 0.064           | 0.069 |                                                      | 0.047           | 0.062 |                                        | 0.054           | 0.053 |

<sup>a</sup> Items added only for the Confirmatory Factor Analyses; \* Items that overlap across all three algorithms; † Items that overlap across two algorithms. Factors that are *not* included in the algorithm cutoffs are *italicized*. C Current; E Ever; 12-20/NV21-47 Children from 12-20 months of age and nonverbal children from 21-47 months of age; SW21-47 Children with single words from 21-47 months of age; PH21-47 Children with phrase speech from 21-47 months of age; EFA Exploratory Factor Analyses; CFA Confirmatory Factor Analyses; CFI Comparative Fit Index; RMSEA Root Mean Square Error Approximation. ASD Autism Spectrum Disorder, NS Nonspectrum disorder, TD Typical Development

other factors for both groups ( $\beta = 0.337$ ,  $p < 0.001$  for “12-20/NV21-47”;  $\beta = 0.236$ ,  $p < 0.05$  for “SW21-47”).

In contrast, Imitation, Gestures, and Play (IGP) did not predict diagnosis for ASD versus NS when the other two

factors and participant characteristics were controlled. For this reason, the final algorithm cutoff score for the “12-20/NV21-47” and “SW21-47” algorithms included only the two domains, SA and RRBs, though the third factor will

remain on algorithm forms in order to allow quantification of these deficits.

Item totals using SA and RRB factors were combined, and the combined scores also predicted diagnosis when the third factor, IGP, and participant characteristics were controlled ( $\beta = 0.232$ ,  $p < 0.001$  for “12-20/NV21-47”;  $\beta = 0.327$ ,  $p < 0.001$  for “SW21-47”). For the children with phrase speech, each domain significantly predicted diagnosis while controlling for the other two factors as well as IQ scores and age ( $\beta = 0.148$ ,  $p < 0.05$  for SC;  $\beta = 0.312$ ,  $p < 0.05$  for RRBs;  $\beta = 0.19$ ,  $p < 0.05$  for RPI). Thus, all three factors (SC, RRBs, and RPI) were included when generating diagnostic thresholds for the “PH21-47” algorithm.

For all three algorithms, all domain scores included in the algorithms for *unique cases* were significantly higher for the ASD sample than the nonspectrum or typically developing groups (See Table 3).

#### Correlations Between Domain Totals and Participant Characteristics

Correlations between domain totals and participant characteristics were examined using *unique cases* only, in order to eliminate effects of repeated participation on domain totals. For the “12-20/NV21-47” group, most of the correlations between the domain totals and participant characteristics did not exceed an  $r$  of 0.5. Nevertheless, for the “12-20/NV21-47” group, the correlation between verbal IQ and the IGP domain was higher ( $r = -0.52$ ,  $p < 0.001$ ), which is one of the reasons why it is not included in the diagnostic algorithm. For the “SW21-47” group, most of the correlations between the domain totals and participant

characteristics did not exceed an  $r$  of 0.4. For the “PH21-47” group, no correlations exceeded an  $r$  of 0.4.

#### Internal Consistency of Algorithm

Internal consistency of each algorithm was examined through correlations between item and domain-total-minus-item correlations (item-total correlation) and Cronbach’s Alpha using *unique cases*. The item, *current: inappropriate facial expression*, always loaded onto the RRB domain. As expected, because we placed the item into the SA domain, the item-total correlation for the particular item was often the lowest among all the items in the same domain. For the younger or nonverbal children receiving the “12-20/NV21-47” algorithm, item-total correlations ranged from 0.65 (*current: inappropriate facial expression*) to 0.81 (*current: social smiling*) for the SA domain; from 0.71 (*ever: hand and finger mannerisms*) to 0.77 (*ever: repetitive use of objects*) for the RRB domain; and from 0.64 (*current: imaginative play*) to 0.82 (*current: instrumental gestures*) for the IGP domain. Cronbach’s alpha was 0.9 for the SA domain; 0.73 for the RRB domain, and 0.87 for the IGP domain, indicating strong internal consistency (Cronbach 1951).

For the older children with single words receiving the “SW21-47” algorithm, item-total correlations for *unique cases* ranged from 0.61 (*current: inappropriate facial expression*) to 0.73 (*current: direct gaze*) for the SA domain; from 0.49 (*ever: unusual preoccupation*) to 0.69 (*ever: repetitive use of objects*) for the RRB domain, and from 0.6 (*current: offering to share*) to 0.78 (*current: instrumental gestures*) for the IGP domain. Cronbach’s alpha showed strong internal consistency for the items in

**Table 3** Mean algorithm domain scores by diagnostic group

| Domain                | 12-20/NV21-47  |               |               | SW21-47       |               | PH21-47       |               |               |
|-----------------------|----------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                       | ASD $n = 263$  | NS $n = 47$   | TD $n = 47$   | ASD $n = 127$ | NS $n = 30$   | ASD $n = 101$ | NS $n = 59$   | TD $n = 20$   |
| Social affect         | 11.8<br>(4.18) | 5.7<br>(4.25) | 1.9<br>(1.73) | 11<br>(4.56)  | 3.4<br>(2.88) | 9.7<br>(4.42) | 5.1<br>(3.77) | 1.4<br>(1.35) |
| RRB                   | 4.8<br>(2.32)  | 2<br>(2.01)   | 0.9<br>(1.24) | 5.5<br>(2.64) | 2.3<br>(1.66) | 5.8<br>(2.99) | 2.9<br>(2.9)  | 0.7<br>(1.09) |
| RPI <sup>a</sup>      | —              | —             | —             | —             | —             | 3<br>(1.76)   | 1.6<br>(1.5)  | 1<br>(1.26)   |
| Total algorithm score | 16.5<br>(5.51) | 7.7<br>(5.25) | 2.8<br>(2.22) | 16.6<br>(6.3) | 5.7<br>(3.72) | 18.5<br>(7)   | 9.5<br>(6.71) | 3.1<br>(2.96) |

Standard deviations in parentheses

12-20/NV21-47 children from 12 to 20 months of age and nonverbal children from 21 to 47 months of age, SW21-47 children with single words from 21 to 47 months of age, PH21-47 children with phrase speech from 21 to 47 months of age, ASD autism spectrum disorder, NS non-spectrum disorder, TD typical development, RRB restricted and repetitive behaviors

<sup>a</sup> Reciprocal and peer interaction domain was included only in the “PH21-47” algorithm

each domain (0.85 for the SA domain; 0.62 for the RRB; 0.74 for the IGP).

For the children with phrase speech receiving “PH21–47” algorithm, item-total correlations ranged from 0.32 (*current: use of other’s body to communicate*) to 0.7 (*current: quality of social overtures*) for the SC domain; from 0.62 (*ever: hand and finger mannerisms*) to 0.76 (*current: stereotyped language*) for the RRBs domain; from 0.81 (*current: appropriateness of social response*) to 0.86 (*current: interest in other children*) for the RPI domain. Cronbach’s alpha showed strong internal consistency for all items in each domain (0.83 for the SC domain; 0.79 for the RRB; 0.72 for the RPI).

### Ranges of Concern

Recognizing that diagnoses of ASD in very young children may be less stable than diagnoses at older ages, *ranges of concern* were identified for all three algorithms to be used for clinical purposes. Three *ranges of concern* were set for each algorithm such that at least 80% of children with ASD and no more than about 5% of children with TD would fall in the two ranges of clinical concern (*mild-to-moderate* and *moderate-to-severe* ranges). See Fig. 1 for results. For all three groups, 67–81% of children with NS were accurately assigned to the *little-to-no* range depending on the developmental group.

### Sensitivity and Specificity of the New Algorithms

We next used ROC curves to generate two sets of cutoffs for the new algorithms; one for clinical purposes with maximum sensitivity and adequate specificity (above 70%) for the comparison of ASD vs. NS and one for research purposes with maximum specificity and adequate sensitivity (above 80%) for the comparison of AUT vs. NS. These cutoffs are tied to the endpoints of the *mild-to-moderate* range from the *ranges of concern* described above.

The clinical cutoffs yielded sensitivities ranging from 80 to 94% and specificities ranging from 70 to 81% for ASD vs. NS depending on developmental cells. For research cutoffs, sensitivities ranged from 80 to 84% and specificities range from 82 to 90% for AUT vs. NS. See Table 4 for more details.

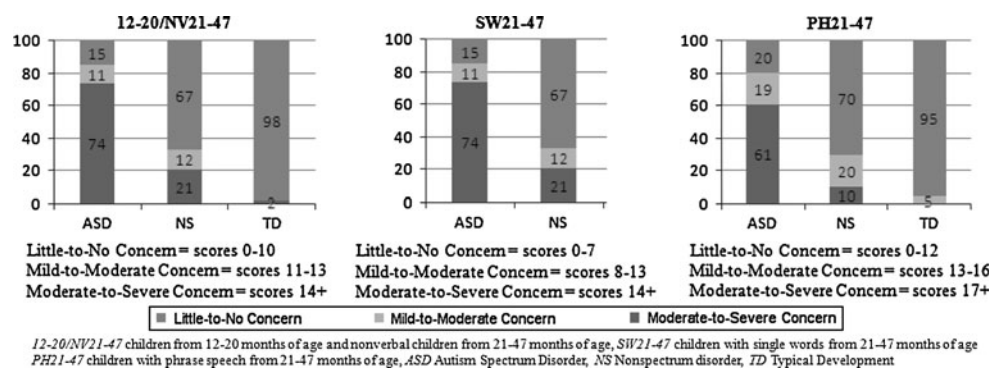
### Comparison of the New Algorithms to the Previous Algorithm

Figure 2 shows the significant gains in predictive validity, using the new algorithms compared to the previous algorithm. As intended, the groups that were harder to differentiate using the previous algorithm showed the most predictive improvement when the new algorithms were used: For the younger and older nonverbal children (“12–20/NV21–47”), specificity improved significantly when either clinical or research cutoffs were used compared to the original algorithm even though sensitivity dropped; for the children with phrase speech (“PH21–47”), specificity and sensitivity improved significantly when either clinical or research cutoffs were used compared to the *current behavior algorithm*. For the older children with single words (“SW21–47”), the new and original algorithms were comparable.

### Discussion

The algorithms presented in this study are the first algorithms developed on data obtained from toddlers to young preschoolers whose ages ranged from 12 to 47 months. These new algorithms offer theoretically updated and more valid ways of using caregiver information in the diagnosis of young children, while expanding the lowest age of application to 12 months with a lowest nonverbal developmental level of 10 months. Compared to the existing algorithms which contain over 30 items, the new algorithms showed improved predictive validity with fewer

**Fig. 1** Percent of participants falling into ranges of concern by diagnostic group



**Table 4** Sensitivity and specificity of research and clinical cutoffs

|               |                      | Sensitivity |           | Specificity |         |
|---------------|----------------------|-------------|-----------|-------------|---------|
|               |                      | AUT         | ASD       | NS          | NS & TD |
| 12-21/NV21-47 | Research cutoff = 13 | 84          | 77        | <b>85</b>   | 93      |
|               | Clinical cutoff = 11 | 91          | <b>85</b> | 70          | 86      |
| SW21-47       | Research cutoff = 13 | 80          | 71        | <b>90</b>   | –       |
|               | Clinical cutoff = 8  | 99          | <b>94</b> | 81          | –       |
| PH21-47       | Research cutoff = 16 | 84          | 70        | <b>82</b>   | 86      |
|               | Clinical cutoff = 13 | 93          | <b>80</b> | 70          | 76      |

AUT autism, ASD autism spectrum disorder, NS nonspectrum disorder, TD typical development, 12-20/NV21-47 children from 12 to 20 months of age and nonverbal children from 21 to 47 months of age, SW21-47 children with single words from 21 to 47 months of age, PH21-47 children with phrase speech from 21 to 47 months of age. Bolded numbers indicate maximized specificities and sensitivities depending on criteria used in selecting cutoff scores

items (13–20 items). In particular, the new algorithms showed substantial gains in specificity (37–42%) for the “12-20/NV21-47” group and modest gain in specificity (2–14%) with consistent improvements in sensitivity (10–14%) for the “PH12-47” group.

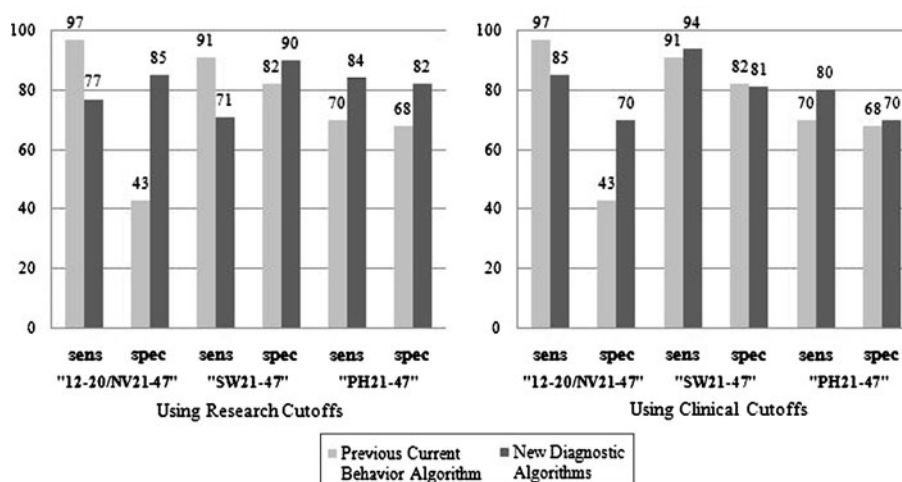
One of the advantages of the new algorithms for toddlers is that they provide clinicians and researchers with several different options for the diagnostic classification of young children. For clinical purposes, *ranges of concern* are proposed that represent the severity of autism symptoms. Depending on where a child falls in among the three *ranges of concern*, a clinician or a researcher can decide about whether or not the child should be followed up with further assessments and enter into treatment irrespective of diagnostic cutoffs. Scores that fall into the *little-to-no* range indicate that the child is reported to have no more behaviors associated with ASD than children in the same age

range who do not have ASD. On the other hand, a child who scores in the *mild-to-moderate* range has a number of behaviors consistent but not unique to ASD. For clinical purposes, these children, just as those in the *moderate-to-severe* range should receive further evaluation and follow-up, including other cognitive and language assessments, and recommendations for treatment.

On the other hand, researchers conducting expensive and time consuming procedures such as neuroimaging may wish to stratify ASD cases in order to more strictly exclude likely NS cases by using the research cutoffs that include only the *moderate-to-severe* range. In contrast, researchers such as geneticists who are casting a broader net for children with autism-related difficulties and clinicians needing to avoid wrongly denying a child access to services can choose to use the clinical cutoffs. In the past, these cutoffs might have been linked to differences between autism and PDD-NOS, but since it is clear that, in this sample, these differences were quantitative, not qualitative, designating them as *ranges of concern* seems more appropriate.

As found in previous research, results from the present study showed that social and communication items primarily loaded into one factor, the *Social Affect* domain for the younger children and older nonverbal children as well as for children with single words and the *Social Communication* domain for children with phrase speech. These results are consistent with past studies using the ADI-R with older children that have also shown that items associated with social and communication loaded onto a single factor (Frazier et al. 2008; Snow et al. 2008; Van Lang et al. 2006). In addition, the present study showed that for all of the three developmental groups, a second factor was associated with RRBs. Items were similar across the groups, but children with phrase speech had additional items such as *stereotyped language* due to their advanced

**Fig. 2** Sensitivities and specificities of new diagnostic algorithms (using research and clinical cutoffs) and a previous current behavior algorithm



Sens, Sensitivity, Spec, Specificity, 12-20/NV21-47 children from 12-20 months of age and nonverbal children from 21-47 months of age, SW21-47 children with single words from 21-47 months of age, PH21-47 children with phrase speech from 21-47 months of age, ASD Autism Spectrum Disorder, NS Nonspectrum disorder, TD Typical Development

verbal abilities. Cronbach's alphas were lower than expected for the RRB domain even though they were all above 0.7, possibly because the domains encompass a diverse set of behaviors. It is also interesting to note that the item, *inappropriate facial expression*, consistently loaded on this domain, raising the possibility that the domain may not only represent RRBs but also unusual behaviors of other types.

Items associated with *Imitation, Play and Gestures* (IGP) loaded onto a third factor for the first two groups of children with minimal language (nonverbal children and children with single words), not for those with phrase speech. This finding is consistent with a recent study done by Frazier et al. (2008) using the items in the ADI-R in which the authors found a third factor related to *Play*. It is interesting that even though the IGP factor consists of "best items" that differentiated children with ASD from those with NS and typically developing children, it did not differentiate between diagnoses when age, IQ scores, and the other domain scores were covaried. This was why we did not incorporate it into the cutoffs for ASD in the "12-20/NV21-47" and "SW21-47" algorithms along with the SA and RRB domains. For the children with phrase speech, the third factor was associated with *Reciprocal and Peer Interaction*, also consistent with past studies (Van Lang et al. 2006). The third factor uniquely contributed to diagnostic differentiation, and it was found to be independent of age and NVIQ, which was why it was included in the algorithm total.

With the new algorithms, children do not have to have RRBs as long as they score high enough on the other domain(s) to exceed the cutoffs for ASD. This may partially ease the concern raised in past studies that parents might not report RRBs in very young children (Wiggins and Robins 2008). Nevertheless, RRB domain totals were consistently higher for children with ASD than children with NS and TD in all of the three developmental cells. Furthermore, in past studies, RRBs had added to stability of diagnoses over time and diagnostic predictability across measures (Lord et al. 2006; Risi et al. 2006). Thus, all of the domains including the RRB domain clearly contributed to the diagnostic validity of the new algorithms.

The goal of creating algorithms less dependent on age was met relatively easily by dividing the sample into cells by age. Minimizing the effect of nonverbal IQ and language level was more complex but low correlations (below 0.4) between each domain score and the participant characteristics were maintained by creating different algorithms for different language levels. For the "12-20/NV21-47" group, verbal IQ scores showed a moderate correlation with the IGP domain even after the sample was divided into different language levels. This is one of the reasons why the IGP domain was not included in the diagnostic

algorithm even though the domain seems sufficiently important to make it readily available on algorithm forms.

## Limitations

Even though we were able to create similar algorithms across the three groups, different thresholds across cells were necessary in order to obtain the best sensitivity and specificity within each developmental cell. This limits the interpretation of data when clinicians and researchers want to measure changes over time because children will move from algorithm to algorithm as they grow older. However, it is not surprising that the algorithms contain slightly different items; some of abnormalities in social interaction and communication as well as RRBs become less or more salient with development. Clinicians can compare items that overlap across algorithms to see the changes in the severity in the specific behaviors measured by each item.

Sensitivity and specificity of the measure may vary in different research samples due to factors such as participant characteristics, socio economic status of the family, and skills of the examiner. In particular, there were few NS children in the 12–20 age group in the present study. However, these children were combined with nonverbal children up to 47 months into the "12-20/NV21-47" group because of similarities in score distributions, and this provided us a sufficient sample size for this group.

Replications across sites with well-defined populations with and without ASD will be critical. For replications, the total scores can be calculated currently by adding scores from the items listed under the first two domains for the "12-20/NV21-47" and "SW21-47" groups and those listed under three domains for the "PH21-47" group (See Table 2). Replications will be needed for each of the two different thresholds for ASD (See Table 1) as well as *ranges of concern* (See Fig. 1).

## Conclusion

In sum, new ADI-R algorithms presented in this study extend the valid use of the ADI-R to toddlers and young preschoolers ranging from 12 to 47 months of age and down to nonverbal mental age of 10 months. Algorithms can be used for either the standard or toddler version of the ADI-R. We hope that researchers and clinicians alike find them a useful tool in supporting families and children with ASD to advance our understanding of these conditions through quantifying autism symptom domains at individual and domain levels and along with clinical observations and other information, contributing to the reliable diagnoses of toddlers and young preschoolers with ASD.

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